Homework 4

Solution

1. Compute each of the following definite integrals

(a)

$$\int_{-1}^{1} \min\left\{e^{x}, e^{-x}, |\mathrm{sign}(x)|\right\}\, dx$$

Solution:

By the patching lemma we have

$$\int_{-1}^{1} \min\left\{e^{x}, e^{-x}, |\mathrm{sign}(x)|\right\}\, dx = \int_{-1}^{0} \min\left\{e^{x}, e^{-x}, |\mathrm{sign}(x)|\right\}\, dx + \int_{0}^{1} \min\left\{e^{x}, e^{-x}, |\mathrm{sign}(x)|\right\}\, dx$$

$$= \int_{-1}^{0} e^{x}\, dx + \int_{0}^{1} e^{-x}\, dx = \left[e^{x}\right]_{-1}^{0} - \left[e^{-x}\right]_{0}^{1} = 1 - \frac{1}{e} - \left(\frac{1}{e} - 1\right) = 2 - \frac{2}{e}$$

<u>Alternative solution:</u>

Note that the function defined by

$$f(x) = \min\left\{e^{x}, e^{-x}, |\mathrm{sign}(x)|\right\} \qquad \forall x \in [-1, 1]$$

is an even function, since

$$f(-x) = \min\left\{e^{-x}, e^{x}, |\,\mathrm{sign}(-x)\,|\right\} = \min\left\{e^{x}, e^{-x}, |\,\mathrm{sign}(x)\,|\right\} = f(x) \qquad \forall x \in [-1, 1]$$

Therefore

$$\int_{-1}^{1} \min\left\{e^{x}, e^{-x}, |\,\mathrm{sign}(x)\,|\right\}\, dx = 2\int_{0}^{1} \min\left\{e^{x}, e^{-x}, |\,\mathrm{sign}(x)\,|\right\}\, dx = 2\int_{0}^{1} e^{-x}\, dx = 2\left[-e^{-x}\right]_{0}^{1} = 2\left(-\frac{1}{e} + 1\right) = 2 - \frac{2}{e}$$

(b)

$$\int_0^1 \ln\left(1+x^2\right)\, dx$$

Solution:

We have

$$\int_0^1 \ln\left(1+x^2\right)\, dx \overset{\text{IBP}}{=} \left[x\cdot\ln\left(1+x^2\right)\right]_0^1 - \int_0^1 \frac{2x^2}{1+x^2}\, dx = \ln(2) - \int_0^1 \frac{2\left(1+x^2\right)}{1+x^2} dx + \int_0^1 \frac{2}{1+x^2}\, dx$$

$$= \ln(2) - 2\int_0^1 dx + 2\int_0^1 \frac{dx}{1+x^2} = \ln(2) - 2\left[x\right]_0^1 + 2\left[\arctan x\right]_0^1 = \ln(2) - 2 + \frac{\pi}{2}$$

(c)

$$\int_0^{\frac{1}{2}} \arccos(x)\, dx$$

Solution:

We use integration by parts,

$$\int_0^{\frac{1}{2}} \arccos(x)\, dx \overset{u'=1,\, v=\arccos(x)}{=} \left[x\cdot\arccos(x)\right]_0^{\frac{1}{2}} + \int_0^{\frac{1}{2}} \frac{x}{\sqrt{1-x^2}}\, dx \overset{t=1-x^2\Rightarrow dt=-2xdx}{=} \frac{1}{2}\arccos\left(\frac{1}{2}\right) - \frac{1}{2}\int_1^{\frac{3}{4}} \frac{dt}{\sqrt{t}}$$

$$= \frac{\pi}{6} + \int_{\frac{3}{4}}^1 \frac{dt}{2\sqrt{t}} = \frac{\pi}{6} + \left[\sqrt{t}\right]_{\frac{3}{4}}^1 = \frac{\pi}{6} + 1 - \frac{\sqrt{3}}{2}$$

We now give an alternative solution:

we use the method of substitution: As $\sin(t) \geq 0$ for every $t \in \left[0, \frac{\pi}{2}\right]$ and as $\cos^2(t) + \sin^2(t) = 1$, then $\sin(t) = \sqrt{1-\cos^2(t)}$ for every $t \in \left[0, \frac{\pi}{2}\right]$.

We now compute

$$\int_0^{\frac{1}{2}} \arccos(x)\, dx \overset{t=\arccos(x)\rightarrow dt=-\frac{dx}{\sqrt{1-x^2}}}{=} -\int_{\frac{\pi}{2}}^{\arccos\left(\frac{1}{2}\right)} t\sqrt{1-x^2}\, dt = \int_{\arccos\left(\frac{1}{2}\right)}^{\frac{\pi}{2}} t\sqrt{1-\cos^2(t)}\, dt$$

$$= \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} t\sin(t)\, dt \overset{u'=\sin(t),\, v=t}{=} \left[-t\cos(t)\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos(t)\, dt$$

$$= \frac{\pi}{6} + \int_{\frac{\pi}{3}}^{\frac{\pi}{2}} \cos(t)\, dt = \frac{\pi}{6} + \left[\sin(t)\right]_{\frac{\pi}{3}}^{\frac{\pi}{2}} = \frac{\pi}{6} + 1 - \frac{\sqrt{3}}{2}$$

2. (a) Compute the definite integral $\int_0^{\frac{\pi}{4}} \tan(x)\,dx$.

Solution:

We have

$$\int_0^{\frac{\pi}{4}} \tan(x)\,dx = \int_0^{\frac{\pi}{4}} \frac{\sin(x)}{\cos(x)}\,dx \overset{t=\cos(x)\Rightarrow dt=-\sin(x)dx}{=} -\int_1^{\frac{2}{\sqrt{2}}} \frac{dt}{t} = -\left[\ln|t|\right]_1^{\frac{\sqrt{2}}{2}} = -\ln\left(\frac{\sqrt{2}}{2}\right) = \frac{1}{2}\ln(2)$$

(b) Conclude that

$$\frac{\ln(2)}{2e} \leq \int_0^{\frac{\pi}{4}} e^{-x}\tan(x)\,dx \leq \frac{\ln(2)}{2}$$

Note: you may assume that $\pi < 4$.

Solution:

Note that the function $e^{-x}\tan(x)$ is continuous on $\left[0, \frac{\pi}{4}\right]$ by AOC, and therefore the definite integral $\int_0^{\frac{\pi}{4}} e^{-x}\tan(x)\,dx$ exists. Consider the function defined by $f(x) = e^{-x}$ for every $x \in \left[0, \frac{\pi}{4}\right]$.

Note that f is decreasing on $\left[0, \frac{\pi}{4}\right]$ and therefore

$$\frac{1}{e^{\frac{\pi}{4}}} = f\left(\frac{\pi}{4}\right) \leq f(x) \leq f(0) = 1 \qquad \forall x \in \left[0, \frac{\pi}{4}\right]$$

multiplying the above inequality by $\tan(x) \geq 0$ for every $x \in \left[0, \frac{\pi}{4}\right]$ we obtain

$$\frac{\tan(x)}{e^{\frac{\pi}{4}}} \leq e^{-x}\tan(x) \leq \tan(x) \qquad \forall x \in \left[0, \frac{\pi}{4}\right]$$

Integrating all sides of the inequality, then by the monotonicity of the definite integral the inequalities are preserved. Using item (a) we conclude that

$$\frac{\ln(2)}{2e} \overset{\pi<4}{\leq} \frac{\ln(2)}{2e^{\frac{\pi}{4}}} = \frac{1}{e^{\frac{\pi}{4}}}\int_0^{\frac{\pi}{4}} \tan(x)\,dx \leq \int_0^{\frac{\pi}{4}} e^{-x}\tan(x)\,dx \leq \int_0^{\frac{\pi}{4}} \tan(x)\,dx = \frac{\ln(2)}{2}$$

3. Let $a, b \in \mathbb{R}$ such that $a < b$ and let $f : [a, b] \to \mathbb{R}$ be a function that is integrable on $[a, b]$. Let $x_0 \in \mathbb{R}$, let $\phi : \mathbb{R} \to [a, b]$ and let $G : \mathbb{R} \to \mathbb{R}$ be the function that is defined by $G(x) = \int_a^{\phi(x)} f(s)\, ds$ for every $x \in \mathbb{R}$.

(a) Prove that G is well defined.

Solution:

As $\phi : \mathbb{R} \to [a, b]$, then $\phi(x) \in [a, b]$ for every $x \in \mathbb{R}$. As f is integrable on $[a, b]$, then f is integrable on $[a, \phi(x)]$ for every $x \in \mathbb{R}$ and thus, G is well defined in $\mathbb{R}$.

(b) Prove that if f is continuous at $y_0 = \phi(x_0)$ and if ϕ is differentiable at x_0, then G is differentiable at x_0 and $G'(x_0) = f(\phi(x_0)) \cdot \phi'(x_0)$.

Solution:

Consider the function that is defined by $h(y) = \int_a^y f(s)\, ds$ for every $y \in [a, b]$. As f is continuous at $\phi(x_0)$, then by the Fundamental theorem of calculus h is differentiable at $\phi(x_0)$ and $h'(\phi(x_0)) = f(\phi(x_0))$. Note that $G(x) = h(\phi(x)) = \int_a^{\phi(x)} f(s)\, ds$ for every $x \in \mathbb{R}$. By the chain rule, G is differentiable at x_0 and

$$G'(x_0) = h'(\phi(x_0)) \cdot \phi'(x_0) = f(\phi(x_0)) \cdot \phi'(x_0)$$

(c) Compute the limit

$$\lim_{x \to 0^+} \frac{\int_0^{x^2} \ln\left(1 + \sqrt{t}\right) dt}{x^3}$$

Solution:

Consider the function defined by $G(x) = \int_0^{x^2} \ln\left(1 + \sqrt{t}\right) dt$ for every $x \geq 0$ .We define $f(x) = \ln\left(1 + \sqrt{x}\right)$ for every $x \geq 0$. By AOC f is continuous. Also let us define $\phi(x) = x^2$ for every $x \geq 0$. By AOD ϕ is differentiable for every $x \geq 0$ and $\operatorname{Im}(\phi) = \{x : x \geq 0\}$. By item (b) G is differentiable at $x_0 = 0$, therefore continuous at $x_0 = 0$, so $\lim_{x \to 0^+} G(x) = G(0)$. We note that

$$G(0) = \int_0^{0^2} \ln\left(1 + \sqrt{t}\right) dt = \int_0^{0} \ln\left(1 + \sqrt{t}\right) dt = 0$$

and therefore

$$\lim_{x \to 0^+} G(x) = G(0) = 0$$

We now compute the limit

$$\lim_{x \to 0^+} \frac{\int_0^{x^2} \ln\left(1 + \sqrt{t}\right) dt}{x^3} \overset{\frac{0}{0}}{=} \lim_{x \to 0^+} \frac{\ln\left(1 + \sqrt{x^2}\right) \cdot 2x}{3x^2} \overset{x>0 \Rightarrow \sqrt{x^2}=x}{=} \lim_{x \to 0^+} \left[\frac{2}{3} \cdot \frac{\ln(1+x)}{x}\right] \overset{\frac{0}{0}}{=} \lim_{x \to 0^+} \frac{2}{3(1+x)} = \frac{2}{3}$$

4. Prove that

$$\lim_{n\to\infty}\left[\frac{1}{n}\sum_{k=1}^{n}\frac{1}{1+\left(\frac{k}{n}\right)^2}\right]=\frac{\pi}{4}$$

Solution:

Consider the function defined by

$$f(x)=\frac{1}{1+x^2}\qquad\forall x\in[0,1].$$

For every $n\in\mathbb{N}$, consider the uniform partition of $[0,1]$ of size n defined by

$$P_n=(x_0,\ldots,x_n)\qquad\text{where}\qquad x_k=\frac{k}{n},\ \forall 0\le k\le n.$$

Note that

$$\Delta(P_n)=\max\left\{x_k-x_{k-1}:1\le k\le n\right\}=\max\left\{\frac{1}{n}\right\}=\frac{1}{n},$$

and therefore

$$\lim_{n\to\infty}\Delta(P_n)=0.$$

Also, for every $x,y\in[0,1]$,

$$|f(x)-f(y)|=\left|\frac{1}{1+x^2}-\frac{1}{1+y^2}\right|=\left|\frac{y^2-x^2}{(1+x^2)(1+y^2)}\right|\le\left|y^2-x^2\right|\le 2\,|x-y|\,.$$

Hence f is Lipschitz on $[0,1]$.

Now note that f is decreasing on $[0,1]$, since for every $x\in[0,1]$

$$f'(x)=\frac{-2x}{(1+x^2)^2}\le 0.$$

Therefore, for every $1\le k\le n$,

$$m_k=\inf\left\{f(x):x_{k-1}\le x\le x_k\right\}=f(x_k)=\frac{1}{1+\left(\frac{k}{n}\right)^2}.$$

By HW3Q6, we conclude that

$$\begin{aligned}\lim_{n\to\infty}\left[\frac{1}{n}\sum_{k=1}^{n}\frac{1}{1+\left(\frac{k}{n}\right)^2}\right]&=\lim_{n\to\infty}\sum_{k=1}^{n}\left[\frac{1}{1+\left(\frac{k}{n}\right)^2}\cdot\frac{1}{n}\right]\\&=\lim_{n\to\infty}\sum_{k=1}^{n}m_k\cdot(x_k-x_{k-1})\\&=\lim_{n\to\infty}\underline{S}(P_n,f)\\&=\int_0^1\frac{dx}{1+x^2}=[\arctan x]_0^1=\frac{\pi}{4}.\end{aligned}$$